Bihar Government Hospital Grievance Redressal System

A QR-code-based, direct-to-government complaint system for patients — designed for AIIMS Patna as a pilot reference, built to scale statewide.

System Design & Implementation Brief

1. The Problem

Patients at government hospitals in Bihar face common, recurring issues — doctors absent from duty, staff not coordinating with patients, rude behavior, medicines unavailable, and more. Existing complaint channels (hospital helpdesks, complaint numbers) route complaints back through the same hospital hierarchy that is often the source of the problem. Complaints get buried, no action is taken against staff who ignore their duty to respond, and patients — many of whom are not highly literate or comfortable with lengthy digital forms — often don't bother complaining at all.

What government wants

- Complaints should reach a government authority **directly**, not filtered through the hospital's own staff.
- If the person responsible for handling complaints (the officer) also fails to act, that inaction itself must be visible and escalated to a higher authority, with the officer's identity attached.
- The process for a patient to file a complaint must be extremely simple — no long forms, no login, minimal literacy required.

2. Core Idea

Every patient at a government hospital already gets a physical file created with their details (name, department, mobile number, ABHA number) — this happens today at AIIMS Patna and similar hospitals. We attach a **QR code** to that same file. Scanning it opens a chat (Telegram for the MVP demo; WhatsApp Cloud API once approved) where the patient's identity is already known, and they simply describe their problem — by typing or by voice note. No form, no category menu, no login.

The complaint is automatically classified by AI (category, urgency, sentiment, one-line summary) and routed to a **government-appointed grievance officer** — someone who is explicitly NOT hospital staff, so hospital administration cannot suppress or influence how the complaint is handled. If that officer fails to act within a defined time limit, the complaint and the officer's inaction are automatically escalated to a state-level superadmin.

3. Why a QR Code + Chatbot (Not a Website Form)

- **No new app to install** for the pilot — Telegram is free and requires no approval; production target is WhatsApp Cloud API, which almost everyone in Bihar already has installed.
- **No typing personal details** — the QR resolves identity automatically via a secure token, so patients (many with limited literacy or comfort with forms) skip the hardest part entirely.
- **Voice-first option** — a patient can simply speak their complaint in Hindi, Bhojpuri, Maithili, or mixed language; speech-to-text handles the rest.
- **One QR works across visits** — printed once on the patient's file/health card, reusable for every future hospital visit, rather than being regenerated and reprinted each time.

4. How the QR Code Works (Identity Without Login)

The QR code does not contain the patient's actual personal details. It encodes a link containing a long, random, unique **token** (e.g. `a8f3k9x2`). This token is stored in the backend database, linked to the patient's real record (name, mobile, ABHA number, hospital, department). When scanned, the messaging app opens a chat with the token pre-filled as the starting message. The bot looks up the token, retrieves the patient's details instantly, and confirms identity back to the patient with a simple Yes/No — no typing, no login screen, no password.

Why not put personal data directly in the QR? If someone photographs or copies the QR/token, they would only get an opaque ID, not the patient's actual name, mobile number, or ABHA number — those stay safely in the backend database, accessible only through the authenticated bot flow.

The same QR is reused for every future visit, since it is tied to the **patient**, not to a single hospital visit. If a patient has visited multiple departments and it's unclear which one a new complaint relates to, the bot asks one short, button-based follow-up question — never open-ended typing.

5. Step-by-Step Complaint Flow

1. Patient (or attendant) scans the QR code on the hospital file.
2. Chat opens automatically; bot resolves the token and confirms identity ('Are you Ramesh Kumar, Cardiology, AIIMS Patna? Yes/No').
3. Bot asks the patient to describe the problem — by typing or sending a voice note. This is the **only** input step required from the patient.
4. If voice: the audio is transcribed to text using a speech-to-text model tuned for Indian languages. The original audio file is always kept alongside the transcript.
5. The complaint text is sent to an AI model, which returns: a category (from a flexible, evolving tag list — not a fixed dropdown the patient has to pick from), a sentiment reading, an urgency level (critical / high / normal / low), and a one-line plain-language summary.
6. The system generates a unique, human-readable complaint ID (e.g. `BH-AIIMSPAT-20260705-0042`) and confirms filing to the patient immediately, along with the AI-assigned category — giving them a chance to flag it if mislabeled.
7. The complaint is routed to the appropriate government grievance officer based on hospital/district, and SLA timers start.

Why AI decides the category (not the patient)

Earlier designs considered asking the patient to pick a complaint category from a menu. This was removed to keep the interaction to a single message with zero extra steps. Instead, an AI model reads the free-form complaint (text or transcribed voice) and infers the category, sentiment, and urgency automatically. The category list itself is not hardcoded — it lives in the database as an editable set of tags. If the AI encounters a complaint that doesn't fit any existing tag well, it proposes a new tag, which a superadmin later reviews and approves or merges with an existing one. This lets the taxonomy grow organically from real complaints rather than being guessed upfront, while avoiding duplicate near-identical tags (e.g. 'doctor absent' vs 'doctor not present').

If classification is uncertain or fails outright, the system defaults to treating the complaint as high-urgency rather than risking it being under-prioritized — the raw text and audio are always preserved and shown to the officer regardless of what the AI concludes.

6. Roles in the System

Patient

No account, no login, no password. Identity is resolved via the QR token only. Can check status anytime by messaging the bot with their complaint ID, and can reply within the same chat thread if an officer reaches out. Can optionally mark a complaint as anonymous, hiding their identity from the officer while remaining fully traceable to the superadmin for audit purposes (protects against retaliation fears for sensitive complaints like staff misbehavior).

Grievance Officer — Government-appointed, NOT hospital staff

This is the central structural fix. Officers are appointed by the government and assigned to a district or hospital cluster (e.g. 'all government hospitals in Patna district') — they are not employees of any hospital they monitor. This independence prevents hospital administration from pressuring officers to bury or downplay complaints about their own staff.

Officer dashboard shows:

- A queue of assigned complaints, each displaying the AI-assigned category in bold and a one-line summary at the top — full detail (raw text, audio) available on click.
- A single toggle to mark a complaint 'Received / Acknowledged' — this instantly notifies the patient through the same chat thread they complained in.
- An in-thread real-time chat option to talk to the patient directly, exactly like a normal messaging conversation, if more information or explanation is needed.
- A 'Mark Resolved' action, which notifies the patient and triggers an automatic follow-up question to the patient: 'Was your issue actually resolved?' If the patient says no, the complaint reopens and escalates automatically — officers cannot self-certify resolution.

Superadmin — State health department level

Oversees all officers, all hospitals, and all complaints statewide. Can create new officer accounts (name, email, resettable password), deactivate or terminate officers, and reassign hospital/district clusters between officers.

Superadmin dashboard shows (kept visually simple for non-technical presentation):

- Total complaints, pending vs resolved counts, average resolution time.
- Complaint volume broken down by category and by hospital — 'what kind of complaints are most common, and from where.'
- Per-officer performance: acknowledgment rate, average resolution time, and number of SLA breaches — so patterns of neglect are visible, not just isolated incidents.
- Every escalated complaint, shown with full context: the original complaint, the patient's detail, and the responsible officer's identity and breach history.

7. Response Timelines & Escalation Logic

Two separate clocks are tracked for every complaint, because an unacknowledged complaint is itself a warning sign, independent of whether it eventually gets resolved.

Stage	Standard SLA	Critical-urgency SLA	What happens on breach
Acknowledgment	30 minutes	15 minutes	Immediately flagged to superadmin dashboard as an acknowledgme
Resolution	48 hours (from acknowledgment)	4 hours	Auto-escalated to superadmin with the full complaint, patient detail,

- The escalation chain is configurable, not fixed at just two tiers — additional levels (e.g. a district health authority above the superadmin) can be added later without redesigning the system.
- If a patient replies 'No' to the post-resolution check ('was this actually fixed?'), that is treated as an automatic reopen-and-escalate event, logged the same way as a formal SLA breach.
- 'Critical' urgency (assigned by the AI classifier) is reserved for situations with potential immediate patient-safety risk — e.g. no doctor available during an emergency — and carries a tightened SLA.

8. Data Model (Summary)

The full schema is provided as a ready-to-use SQL file (`database/schema.sql`) in the project folder. Core entities:

- **hospitals** / **departments** — master data for each facility.
- **patients** — includes the persistent `patient_token` used for QR-based identity resolution.
- **complaint_tags** — the soft, editable category taxonomy; new tags can be proposed and later approved or merged.
- **officers** — government-appointed grievance officers and superadmins, assigned by district/cluster.
- **complaints** — the core record: raw text, raw audio, AI classification output, status, SLA deadlines, resolution confirmation.
- **complaint_messages** — full audit trail of every message exchanged between patient and officer.
- **escalations** — a complete history of every escalation event per complaint, with reason and responsible party.

9. Technology Stack (Free-Tier, MVP-Ready)

Component	Choice	Why
Messaging channel	Telegram Bot API (demo) → WhatsApp Cloud API (pilot)	Telegram is free with zero approval for demo; WhatsApp Cloud API has a gen
Backend	Node.js or Python	Hosted free on Render/Railway free tier.
Database	Supabase (Postgres)	Free tier includes auth and realtime — useful for a live-updating officer dashbo
AI classification	Google Gemini (free tier)	Handles category/sentiment/urgency/summary generation from complaint text

Speech-to-text	AI4Bharat open models (or open-source Whisper)	Built specifically for Indian languages/dialects; free and self-hostable.
QR generation	Open-source QR library	No cost, simple integration.
Dashboards	React, hosted on Vercel	Free hosting, fast to iterate on for officer/superadmin UI.
SLA scheduling	Cron job or free-tier task queue (e.g. Upstash)	Periodically checks ack/resolution deadlines and fires escalation events.

10. Additional Considerations

- **Non-smartphone access:** some patients/attendants will have no smartphone or data access at all. A printed missed-call number or SMS short-code alongside the QR can cover this gap, using low-cost SMS/IVR gateways.
- **Data privacy (DPDp Act, 2023):** this system handles sensitive personal and health data. A basic consent capture step and minimal data retention are recommended even at MVP stage.
- **Alignment with existing frameworks:** positioning this as a faster, hospital-specific feeder into Bihar's existing Right to Public Grievance Redressal Act framework and the national CPGRAMS system — rather than a competing, parallel system — will make government adoption easier.
- **Abuse protection:** basic rate-limiting per token/mobile number prevents the officer queue from being flooded by spam or prank complaints.
- **Sentiment vs category:** these are tracked as two separate signals — category routes the complaint (what the issue is), while sentiment/urgency helps prioritize within a category (how urgently it needs attention).

11. Suggested Build Order

1. Set up the database using the provided schema.
2. Build the Telegram bot: QR-token identity resolution → complaint intake (text + voice) → speech-to-text → AI classification → store complaint → confirm to patient with complaint ID.
3. Build the officer dashboard: complaint queue, acknowledge toggle, resolve action, in-thread chat.
4. Build the SLA scheduler: a background job checking acknowledgment/resolution deadlines and firing escalation events.
5. Build the superadmin dashboard: officer management, cross-hospital analytics, escalation view.
6. Build the QR generation utility for onboarding new patients and hospitals.

This document accompanies a working project folder (backend scaffolding, database schema, and AI classification code already written) and a separate, ready-to-paste implementation prompt for completing the remaining application.